



White above 4 kHz; and white everywhere once the tones are taken out. A 900 Hz-wide band at 4 kHz averages a reflection out; a 58 Hz band at 250 Hz holds one cancellation, so a tone shows it. With every tone notched, the broadband polar is round to ± 1.3 dB in 92 % of cells from 125 Hz up. **This is not a conclusion to discard everything below 4 kHz:** broadband directivity is usable from 125 Hz now; only blade-tone levels below ~1 kHz need the loudspeaker check, the speed ladder or the measured correction first (remedies report \$00, \$03).

[illegible]

1. Take 811-1892 off the arc.
2. Put a small loudspeaker on the stand at the hub. Play a 20 Hz–20 kHz sine sweep and record all microphones; each microphone's own impulse response gives its echo delays (no clock sync needed), times 343 m/s, the extra paths. Then play tones at 216, 238 and 258 Hz: if the ± 8 dB pattern of Fig. 1 comes back, it is the room; if not, it is the propeller's inflow.
3. Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
4. Run a ladder of 12–15 motor speeds (BPF every ~ 5 Hz, 205–265 Hz) and average the tone polar over it; cover or move the surfaces the sweep names; rerun the fit. Foam works from 650 Hz up; at 250 Hz the levers are averaging, correction and distance, then resonant absorbers.

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.